

AN EXAMPLE OF HOMEWORK AND EXAM MARKING GUIDE

GENERAL CHEMISTRY

Hanoi University of Science and Technology – TROY Program

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Question 1. A glass vessel calibrated to contain 70 oz of water at 4°C was found to weigh 7.03 lbs when empty and dry. Filled with a sodium chloride solution at the same temperature, it was found to weigh 11.99 lbs. Calculate the solution's density in g/mL.

Criteria	Points
Summary of the given information: Vessel's volume: 70 oz Vessel's weight: 7.03 lbs Vessel and sodium chloride solution's weight: 11.99 lbs	0.25
Restatement of the expected answer: density of sodium chloride in g/mL	0.25
Conversion factors and equation: 1 oz / 29.57 mL 1 lbs / 453.6 g Equation: Density = mass/volume	0.25
Weight of solution = 11.99 lbs – 7.03 lbs = 4.96 lbs	0.25
Volume of the solution = Volume of the vessel = 70 oz	0.25
Density of solution = $\frac{4.96 \text{ lbs}}{70 \text{ oz}} \times \frac{1 \text{ oz}}{29.57 \text{ mL}} \times \frac{453.6 \text{ g}}{1 \text{ lbs}} = \boxed{1.08 \text{ g/mL}}$ (3 significant figures, boxed)	0.5

Question 2. The specific heat of the glass used in thermometer manufacture is 0.84 Jg⁻¹K⁻¹. Calculate the heat in calories released to the surrounding when a 10-gram thermometer decreases its temperature from 39 degrees Celsius to 37 degrees Celsius.

Criteria	Points
Summary of the given information: Mass: m = 10 g Specific heat: C = 0.84 Jg ⁻¹ K ⁻¹ Initial temperature: t _{initial} = 39 °C Final temperature: t _{final} = 37 °C	0.25
Restatement of the expected answer: released heat in calories	0.25
Conversion factors and equations: 1 cal / 4.184 J K = 273.15 + C H = CmΔt	0.25
Temperature change: Δt = t _{final} - t _{initial} = (37 + 273.15) – (39 + 273.15) = - 2 K Released heat = H = $\frac{0.84 \text{ J}}{\text{gK}} \times \frac{1 \text{ cal}}{4.184 \text{ J}} \times 10 \text{ g} \times (-2 \text{ K}) = \boxed{-4.0 \text{ cal}}$ (negative, 2 significant figures, boxed)	0.5 0.5

Question 3. Analysis of 4.200 g of vitamin C (ascorbic acid) yields 1.720 g of carbon, 0.190 g of hydrogen, and 2.290 g of oxygen. Calculate the mass percentage of each element in the compound. Should there be other elements in ascorbic acid?

Criteria	Points
Summary of the given information: Ascorbic acid: $m = 4.200 \text{ g}$ C: $m_C = 1.720 \text{ g}$ H: $m_H = 0.190 \text{ g}$ O: $m_O = 2.290 \text{ g}$	0.25
Restatement of the expected answer: mass percent of each element, % other elements in ascorbic acid	0.25
Equation: Mass percent = $100\% \times \text{mass of element} / \text{total mass}$	0.25
Mass percent of carbon in ascorbic acid = $100\% \times \frac{1.720 \text{ g}}{4.200 \text{ g}} = 40.95 \%$	0.25
Mass percent of hydrogen in ascorbic acid = $100\% \times \frac{0.190 \text{ g}}{4.200 \text{ g}} = 4.52 \%$	0.25
Mass percent of oxygen in ascorbic acid = $100\% \times \frac{2.290 \text{ g}}{4.200 \text{ g}} = 54.52 \%$	0.25
The total mass of C, H and O = $m_C + m_H + m_O = 1.720 + 0.190 + 2.290 = 4.200 \text{ g} = m$ There is no other elements in ascorbic acid	0.5

Question 4: An ion was found to contain 13 protons, 15 neutrons, and 10 electrons.

- (a) Write a symbol that represents the given information.
 (b) What if 2 protons and 5 electrons are added to the ion?

(a)	
Summary of the given information: $p = 13, n = 15, e = 10$	0.1
Restatement of the expected answer: symbol that represent the ion	0.1
Atomic number = $p = 13$	0.1
Atomic symbol = Al	0.2
Atomic mass = $n + p = 15 + 13 = 28$	0.2
Charge = $p - e = 13 - 10 = +3$	0.1
The symbol that represents the given information is ${}_{13}^{28}\text{Al}^{3+}$	0.2
(b)	
Summary of the given information: 2 protons and 5 protons added	0.1
Restatement of the expected answer: symbol that represent the newly formed species	0.1
Atomic number = $p + 2 = 15$	0.1
Atomic symbol = P	0.2
Atomic mass = $n + p = 15 + 15 = 30$	0.2
Charge = $p + 2 - (e + 5) = 13 + 2 - (10 + 5) = 0$	0.1
The symbol that represents the ion is ${}_{15}^{30}\text{P}$	0.2

Question 5. Predict whether each of the following compounds is ionic or molecular, based on its formula:

a) BaCl_2

b) SeCl_2

Ba is a metal from group 2A, thus it is ready to donate 2 electrons to fulfill the octet rule Cl is a nonmetal from group 7A, thus it is ready to receive 1 electron or share its electrons with another species to fulfill the octet rule The donating – receiving relationship between Ba and Cl forms an ionic compound. Hence, BaCl_2 is an ionic compound.	1
Se is a nonmetal from group 6A, thus it is ready to receive 2 electron or share its electrons with another species to fulfill the octet rule Cl is a nonmetal from group 7A, thus it is ready to receive 1 electron or share its electrons with another species to fulfill the octet rule Electron sharing between Se and Cl forms a molecular compound. Hence, SeCl_2 is an ionic compound.	1

Note: No point is given to a short statement as “ BaCl_2 is an ionic compound”